



Manajemen Proyek Sistem Informasi

PROJECT MANAGEMENT & IT'S CONTEXT

S1 SISTEM INFORMASI
FAKULTAS REKAYASA INDUSTRI
TELKOM UNIVERSITY





OUTLINE

- Administrasi Perkuliahan
- Proyek & Manajemen Proyek: Apa, Kenapa dan Bagaimana
- Konteks Proyek di Sistem Informasi



Deskripsi Mata Kuliah

- Mata kuliah ini akan membantu dalam menguasai pengetahuan dan keahlian mendasar dalam merencanakan dan mengelola sebuah proyek Sistem Informasi.





COURSE LEARNING OBJECTIVE (CLO)

PLO03-CLO01	Mampu berinisiatif dan bertanggungjawab untuk menyelesaikan berbagai tugas dalam tim
PLO05-CLO02	Mampu menulis dokumen profesional sesuai dengan konteks profesional
PLO08-CLO03	Mampu menggunakan metode atau perangkat lunak tertentu dalam keilmuan sistem informasi dalam konteks penelitian atau proyek nyata
PLO09-CLO01	Mampu memahami prinsip-prinsip manajemen sistem informasi dalam konteks organisasi
PLO09-CLO03	Mampu menyusun perencanaan dalam konteks sistem informasi menggunakan ilmu dan praktek yang relevan





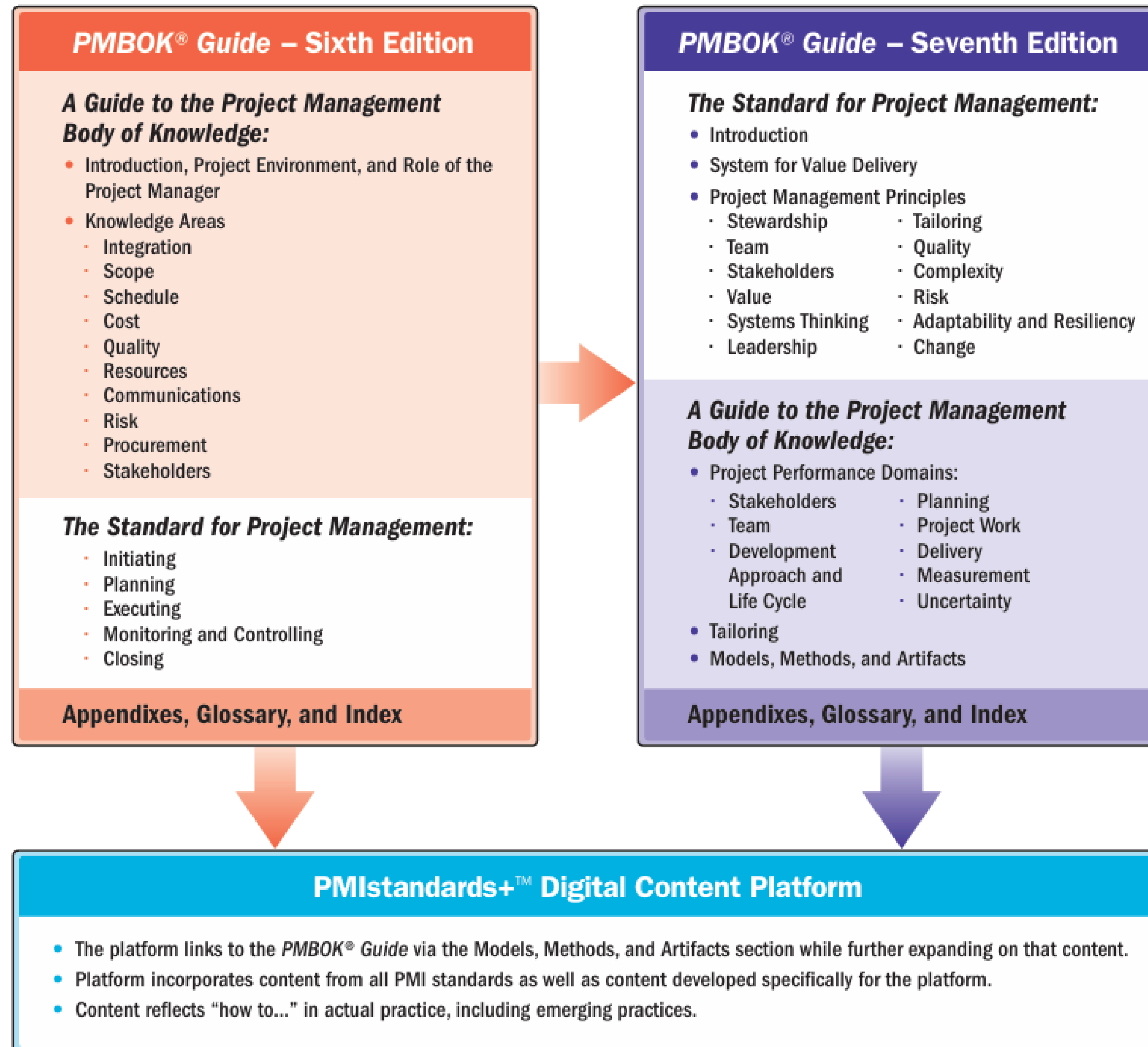
Referensi

Utama:	
1. Project Management Institute. (2017). A Guide to the Project Management Body of Knowledge (6th ed). PMI	
2. Project Management Institute. (2021). A Guide to the Project Management Body of Knowledge (7th ed). PMI	
Pustaka Pendukung:	
1. Schwalbe, Kathy. (2019). Information Technology Project Management (9th ed). Cengage Learning, Inc	

- **PMBOK6** → *Process-based* (berfokus pada **49 proses**, Input-Tools-Techniques-Outputs).
- **PMBOK7** → *Principle-based* (berfokus pada **prinsip & performance domains**, bukan daftar proses detail)



PMBOK 6 → PMBOK 7





Assessment

Id CLO	Bobot per Bentuk Penilaian (%)					TOTAL BOBOT PER SubCLO
	Aktivitas Partisipatif		Kognitif/Pengetahuan		Hasil Proyek	
	Progress Tubes	Penilaian Kontribusi Individu	Quiz	UTS	Tugas Besar	
PLO03-CLO01		10				10
PLO05-CLO02					10	10
PLO08-CLO03	10				15	25
PLO09-CLO01			15	15		30
PLO09-CLO03	10				15	25
Total per penilaian	20	10	15	15	40	100
Utama:						





Nilai Skor Mata Kuliah

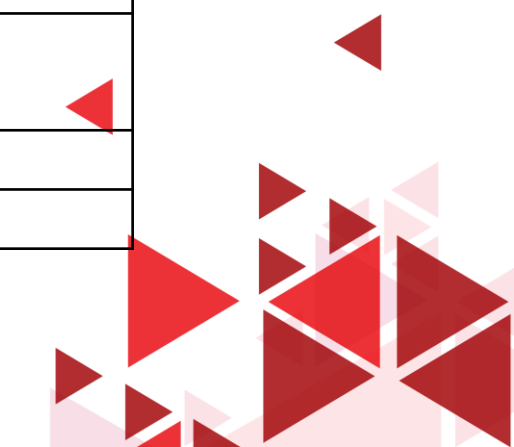
- Nilai Skor Mata Kuliah akan mengikuti aturan terbaru dari Universitas





Rencana Perkuliahan

Pekan ke-	Topik	Assessment	Bobot Nilai	Latihan (Non Assessment)	Deliverables	Referensi	Notes
1	Pengenalan Manajemen Proyek	-	-	Case Base Learning (Pemahaman Terkait Manprosi)		PMBOK6	Pembentukan Kelompok Tugas Besar (Random max 5 orang)
2	Project Management Processes & Project Integration	Quiz-1	1	Case Base Learning (PM Processes)		PMBOK6	Mahasiswa sudah menentukan topik Tugas Besar
3	Project Scope Management	Quiz-2	2	Problem Base Learning (Scope Management Plan & Requirement Management Plan)	Business Case, Statement of Work, WBS, Activity List	PMBOK6	Mahasiswa sudah mulai membuat draft terkait scope sesuai dengan topik tugas besar
4	Project Schedule Management (1)	Quiz-3	2	Problem Base Learning (Schedule)	Gantt Chart, Network Diagram	PMBOK6	
5	Project Schedule Management (2)	-	-	Problem Base Learning (Schedule Lanjutan)		PMBOK6	Mahasiswa sudah mulai membuat draft schedule sesuai dengan topik tugas besar
6	Project Cost Management (1)	Quiz-4	2	Problem Base Learning (Cost)	Project Budget, Funding Limit Requirement, Project S-Curves	PMBOK6	
7	Project Cost Management (2)	-	-	Problem Base Learning (Cost Lanjutan)		PMBOK6	Mahasiswa sudah mulai membuat draft cost sesuai dengan topik tugas besar
8	UTS	UTS	15	Introduction, PM Processess, Scope, Schedule, Cost		PMBOK6	UTS Serentak (PG)
9	Project Resources Management	Quiz-5	2	Problem Base Learning (Resources)	Resources Activity List, RACI Matrix, Resource Requirement	PMBOK6	Mahasiswa Menambahkan Daftar Resource pada bagian Proposal
10	Project Communication Management	Quiz-6	2	Problem Base Learning (Communication)	Communication Management Plan	PMBOK6	Mahasiswa membuat communication management plan sederhana
11	Project Stakeholder Management	Quiz-7	2	Problem Base Learning (Stakeholder)	Stakeholder Register, Stakeholder Analysis	PMBOK6	Mahasiswa membuat stakeholder register sederhana sesuai dengan topik tugas besar
12	Project Risk Management	Quiz-8	2	Problem Base Learning (Risk)	Risk Register	PMBOK6	Mahasiswa membuat list risk management sederhana sesuai dengan topik tugas besar
13	Presentasi Akhir	Peer Review	10	-	Final laporan, Slide Presentasi	PMBOK6	
		Progress Tugas Besar	20				
14	Presentasi Akhir	Progress Tugas Besar			-	Final laporan, Slide Presentasi	PMBOK6
15	Revisi Dokumen	Final Tugas Besar	40		Laporan Hasil Revisi	PMBOK6	Online Asynchronous
16	Revisi Dokumen	Final Tugas Besar			Laporan Hasil Revisi	PMBOK6	Online Asynchronous





Aturan Perkuliahan

- Kelas akan dilakukan secara **onsite** dan **online** selama 1 semester menggunakan LMS
- Ikuti semua peraturan universitas
- Mahasiswa wajib mengikuti semua kegiatan yang ada di LMS atau yang diperintahkan oleh dosen untuk dilakukan
- Ketidakjujuran akademik -> nilai E



Projects & Project Management: What, Why and How





Learning Objective

- Understand the growing **need for better project management**, especially for information technology projects
- Explain **what a project is**, provide examples of projects, attributes of projects, and describe the triple constraint of projects
- Describe **key elements of the project management**, including project stakeholders, the knowledge areas, tools and techniques
- Discuss the **relationship between project, program, and portfolio management** and the contributions to enterprise success
- Understand **the role of the project manager** by describing what project managers do, what skills they need, and what the career field is like
- Describe **the project management profession**, including the role of organizations like the Project Management Institute (PMI) and the importance of certification and ethics





Introduction

- Many organizations today have a new or renewed interest in project management
- Computer hardware, software, networks, and the use of interdisciplinary and global work teams have radically changed the work environment
- The world spends nearly \$10 trillion of its \$40.7 trillion gross product on projects of all kinds
- More than 16 million people regard project management as their profession





Project Management Statistics

- The overall information and communications technology (ICT) market grew by 6 percent to almost **\$3 trillion in 2010**
- In the U.S. the size of the IT workforce topped **4 million workers** for the first time in 2008
- In 2007 the total compensation for the average **senior project manager in U.S. dollars was \$104,776** per year in the United States, \$111,412 in Australia, and \$120,364 in the United Kingdom
- The number of people earning their **Project Management Professional (PMP) certification** continues to increase





Motivation for Studying Information Technology (IT) Project Management

- IT Projects have a **terrible track record**, as described in the What Went Wrong?
- A 1995 Standish Group study (CHAOS) found that **only 16.2% of IT projects were successful** in meeting scope, time, and cost goals; **over 31% of IT projects were canceled** before completion
- A PricewaterhouseCoopers study found that overall, **half of all projects fail** and **only 2.5% of corporations consistently meet their targets** for scope, time, and cost goals for all types of project



Advantages of Using Formal Project Management

- Better **control** of financial, physical, and human resources
- Improved **customer relations**
- Shorter **development times**
- Lower **costs**
- Higher **quality** and increased **reliability**
- Higher **profit margins**
- Improved **productivity**
- Better **internal coordination**
- Higher **worker morale**



What Is a Project?



A project is “a temporary endeavor undertaken to create a unique product, service, or result”



On the other hand, operations is work done to sustain the business



Projects end when their objectives have been reached or the project has been terminated



Projects can be large or small and take a short or long time to complete

Example of IT Project



A technician replaces ten laptops for a small department



A small software development team adds a new feature to an internal software application for the finance department



A college campus upgrades its technology infrastructure to provide wireless Internet access across the whole campus



A cross-functional task force in a company decides what Voice-over-Internet-Protocol (VoIP) system to purchase and how it will be implemented



A government group develops a system to track child immunizations



Project Attributes

- A project:
 - Has a unique **purpose**
 - Is **temporary**
 - Is developed using **progressive elaboration**
 - Requires **resources**, often from various areas
 - Should have a **primary customer** or sponsor
 - The project sponsor usually provides the **direction and funding** for the project
 - Involves **uncertainty**





KEY TERMS AND CONCEPTS (PMBOK7)

- **Outcome** : An end result or consequence of a process or project. Outcomes can include outputs and artifacts, but have a broader intent by focusing on the benefits and value that the project was undertaken to deliver.
- **Portfolio** : Projects, programs, subsidiary portfolios, and operations managed as a group to achieve strategic objectives.
- **Product** : An artifact that is produced, is quantifiable, and can be either an end item in itself or a component item.
- **Program** : Related projects, subsidiary programs, and program activities that are managed in a coordinated manner to obtain benefits not available from managing them individually.
- **Project** : A temporary endeavor undertaken to create a unique product, service, or result. The temporary nature of projects indicates a beginning and an end to the project work or a phase of the project work. Projects can stand alone or be part of a program or portfolio.
- **Project management** : The application of knowledge, skills, tools, and techniques to project activities to meet project requirements. Project management refers to guiding the project work to deliver the intended outcomes. Project teams can achieve the outcomes using a broad range of approaches (e.g., predictive, hybrid, and adaptive).





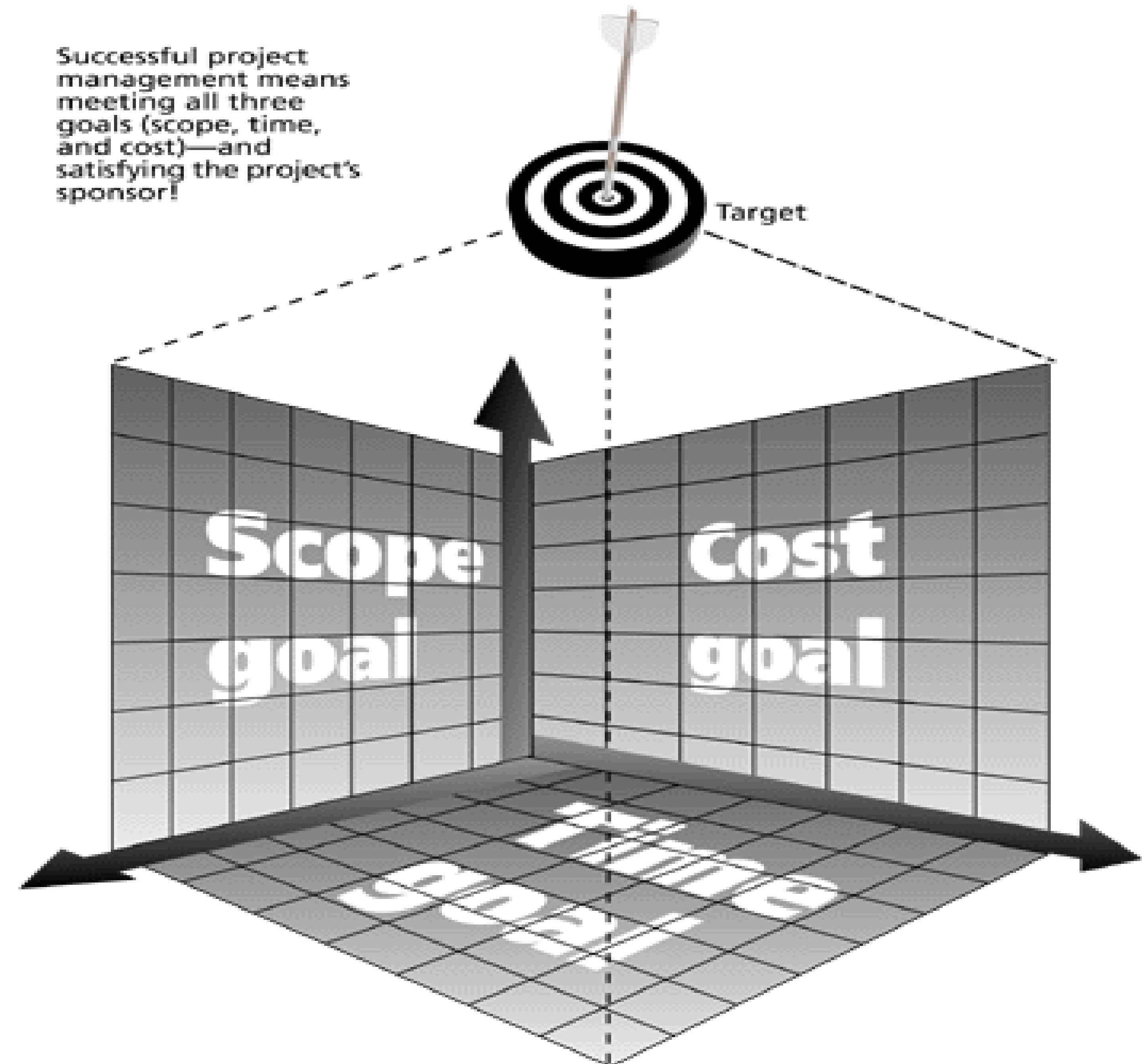
KEY TERMS AND CONCEPTS -2 (PMBOK7)

- **Project manager** : The person assigned by the performing organization to lead the project team that is responsible for achieving the project objectives. Project managers perform a variety of functions, such as facilitating the project team work to achieve the outcomes and managing the processes to deliver intended outcomes.
- **Project team** : A set of individuals performing the work of the project to achieve its objectives.
- **System for value delivery** : A collection of strategic business activities aimed at building, sustaining, and/or advancing an organization. Portfolios, programs, projects, products, and operations can all be part of an organization's system for value delivery.
- **Value** : The worth, importance, or usefulness of something. Different stakeholders perceive value in different ways. Customers can define value as the ability to use specific features or functions of a product. Organizations can focus on business value as determined with financial metrics, such as the benefits less the cost of achieving those benefits. Societal value can include the contribution to groups of people, communities, or the environment.

Triple Constraint of Project Management

Successful project management means **meeting all three goals** (scope, time, and cost)

– and **satisfying the project's sponsor!**





CREATING VALUES (PMBOK7)

- Projects exist within a larger system, such as a governmental agency, organization, or contractual arrangement.
- For the sake of brevity, this standard uses the term organization when referring to government agencies, enterprises, contractual arrangements, joint ventures, and other arrangements. Organizations create value for stakeholders.
 - Creating a new product, service, or result that meets the needs of customers or end users;
 - Creating positive social or environmental contributions;
 - Improving efficiency, productivity, effectiveness, or responsiveness;
 - Enabling the changes needed to facilitate organizational transition to its desired future state; and
 - Sustaining benefits enabled by previous programs, projects, or business operations.





What is Project Management

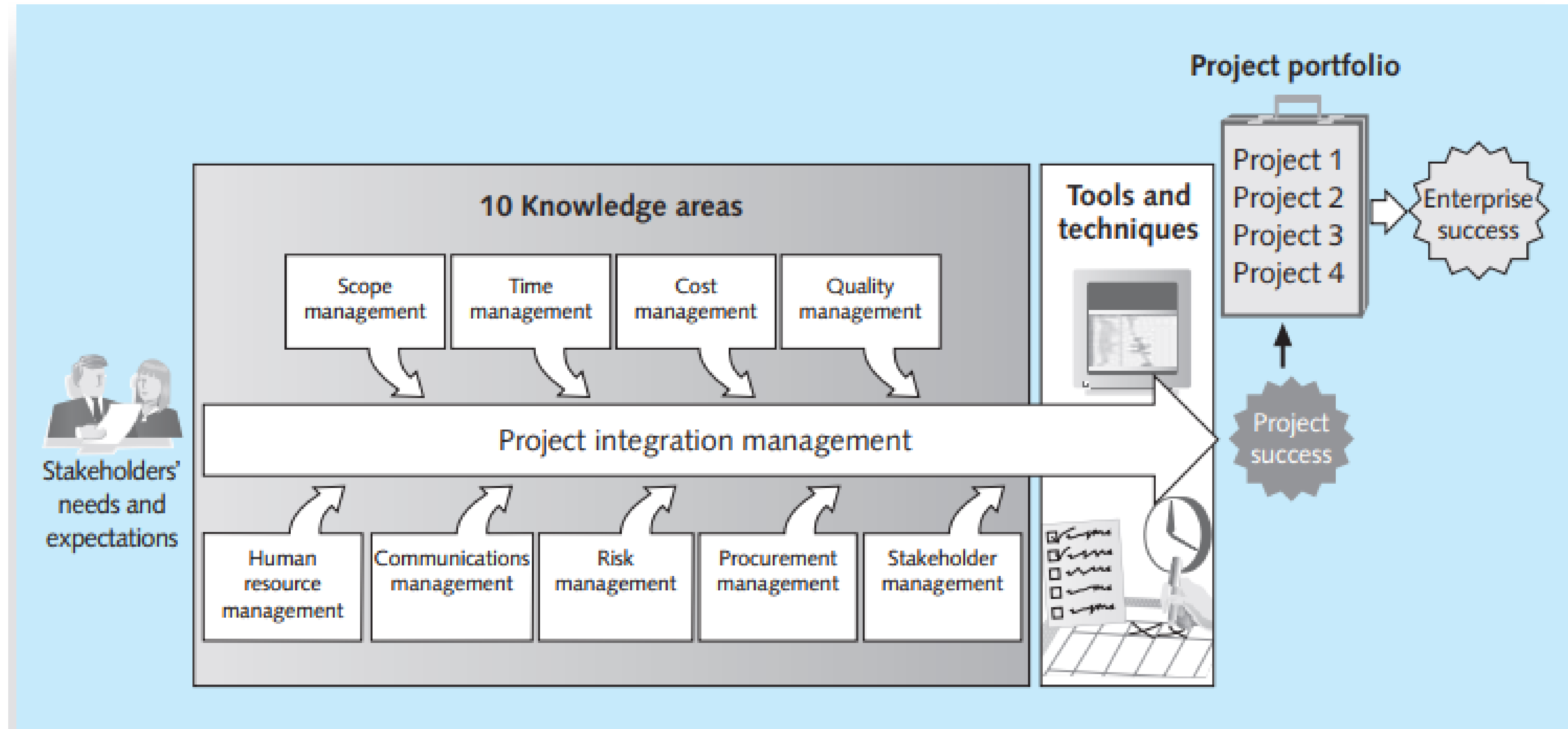


Project management is “the application of knowledge, skills, tools and techniques to project activities to meet project requirements”



Project managers strive to meet the triple constraint by balancing project scope, time, and cost goals

Project Management Framework (PMBOK6)

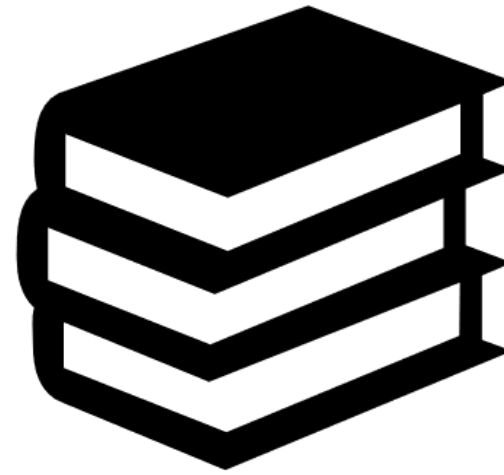




Project Management Framework Component



Stakeholder



Knowledge Areas



Tools &
Techniques





Project Stakeholders

- Stakeholders are the people involved in or affected by project activities
- Stakeholders include:
 - The project sponsor
 - The project manager
 - The project team
 - Support staff
 - Customers
 - Users
 - Suppliers
 - Opponents to the project





10 Project Management Knowledge Areas

- Knowledge areas describe the key competencies that project managers must develop
 - 4 core knowledge areas lead to specific project objectives (scope, time, cost, and quality)
 - 5 facilitating knowledge areas are the means through which the project objectives are achieved (resources, communication, risk, procurement management and stakeholder management)
 - 1 knowledge area (project integration management) affects and is affected by all of the other knowledge areas
- All knowledge areas are important!



Project Management Tools and Techniques

- Project management tools and techniques assist project managers and their teams in various aspects of project management
- Some specific ones include:
 - Project charter, scope statement, and WBS (scope)
 - Gantt charts, network diagrams, critical path analysis, critical chain scheduling (time)
 - Cost estimates and earned value management (cost)
 - See Table on the next slide for many more



Super tools

- “Super tools” are those tools that have high use and **high potential for improving project success**, such as:
 - Software for **task scheduling** (such as project management software)
 - **Scope statements**
 - **Requirements** analyses
 - Lessons-learned reports
- Tools already extensively used that have been found to improve project importance include:
 - **Progress reports**
 - **Kick-off meetings**
 - **Gantt charts**
 - **Change requests**





Project Success

- There are several ways to **define project success**:
 - The project **met scope, time, and cost** goals
 - The project **satisfied the customer/sponsor**
 - The results of the project **met its main objective**, such as making or saving a certain amount of money, providing a good return on investment, or simply making the sponsors happy





What Helps Projects Succeed?

- Executive support
- User involvement
- Experienced project manager
- Clear business objectives
- Minimized scope
- Standard software infrastructure
- Firm basic requirements
- Formal methodology
- Reliable estimates
- Other criteria, such as small milestones, proper planning, competent staff, and ownership



What the Winners Do...

- Recent research findings show that companies that excel in project delivery capability:
 - **Use an integrated project management toolbox** (use standard/advanced PM tools, lots of templates)
 - **Grow project leaders**, emphasizing **business and soft skills**
 - Develop a **streamlined project delivery process**
 - **Measure project health using metrics**, like customer satisfaction or return on investment



Program and Project Portfolio Management

- A program is “a **group of related projects** managed in a coordinated way to obtain benefits and control not available from managing them individually”
- A **program manager** provides **leadership and direction** for the project managers heading the projects within the program
- Examples of common programs in the IT field include **infrastructure, applications development, and user support**





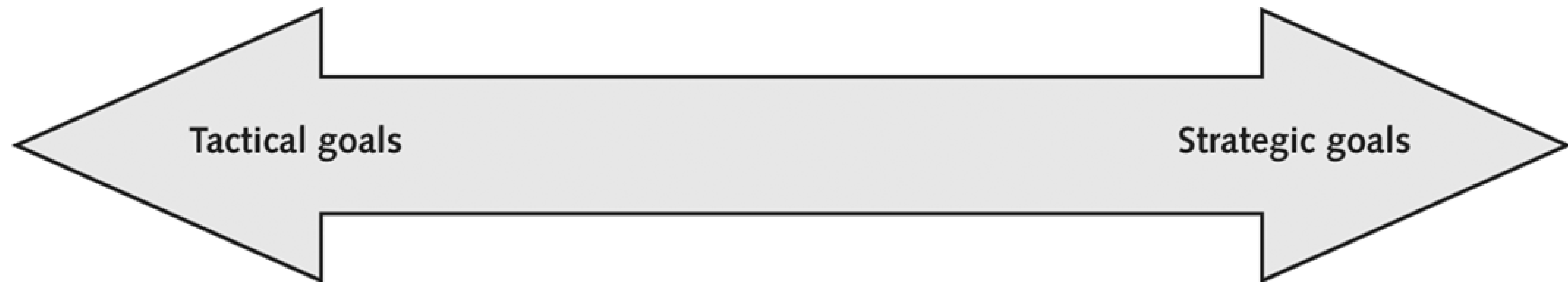
Project Portfolio Management

- As part of project portfolio management, **organizations group and manage projects and programs as a portfolio** of investments that contribute to the entire enterprise's success
- **Portfolio managers** help their organizations make wise investment decisions by helping to **select and analyze projects** from a strategic perspective





Project Vs. Portfolio



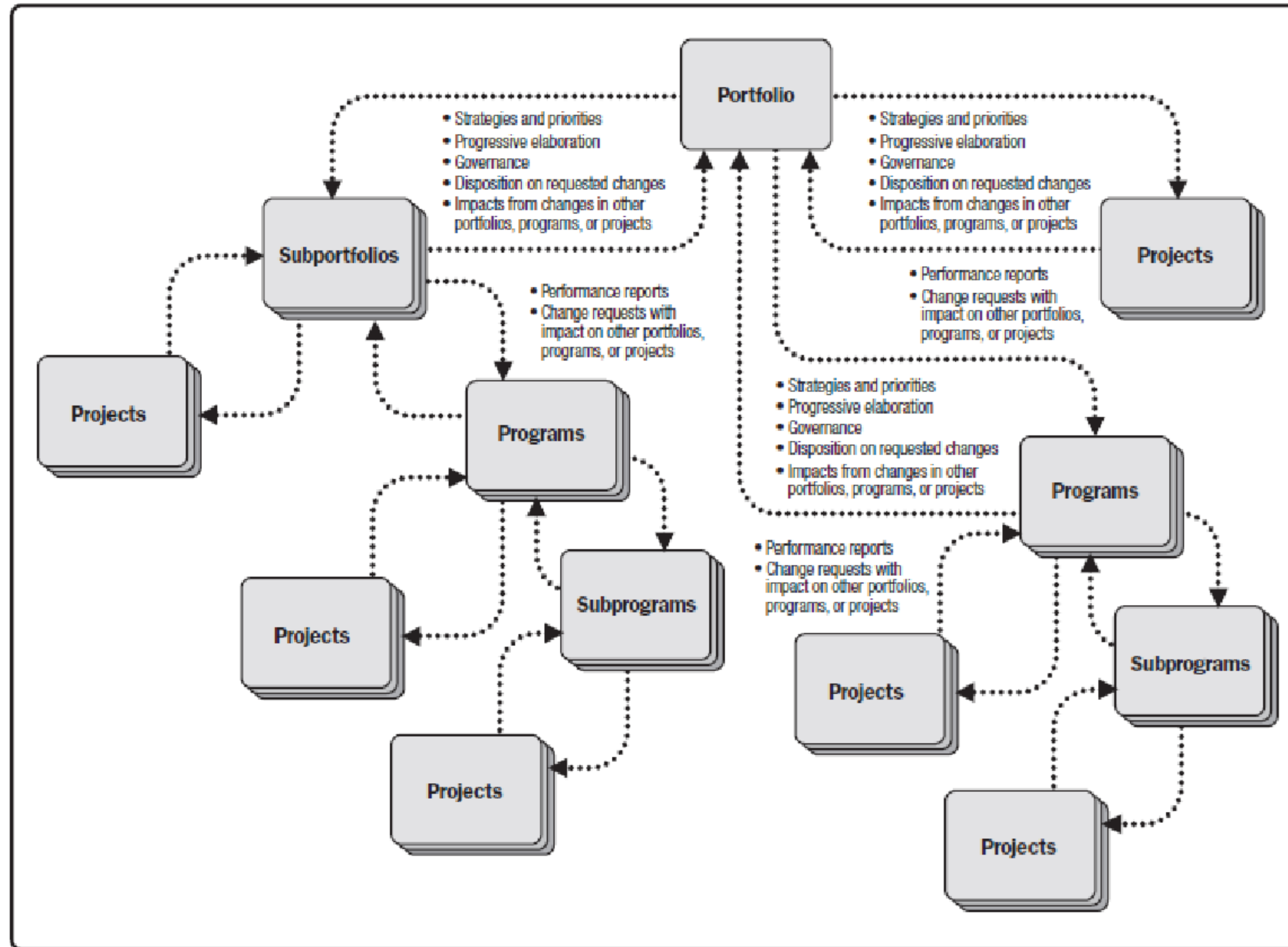
Project management

- Are we carrying out projects well?
- Are projects on time and on budget?
- Do project stakeholders know what they should be doing?

Project portfolio management

- Are we working on the right projects?
- Are we investing in the right areas?
- Do we have the right resources to be competitive?

Project, Program, and Portfolio



VALUE DELIVERY COMPONENTS (PMBOK7)

- A system for value delivery is part of an organization's internal environment that is subject to policies, procedures, methodologies, frameworks, governance structures, and so forth.
- That internal environment exists within the larger external environment, which includes the economy, the competitive environment, legislative constraints, etc.

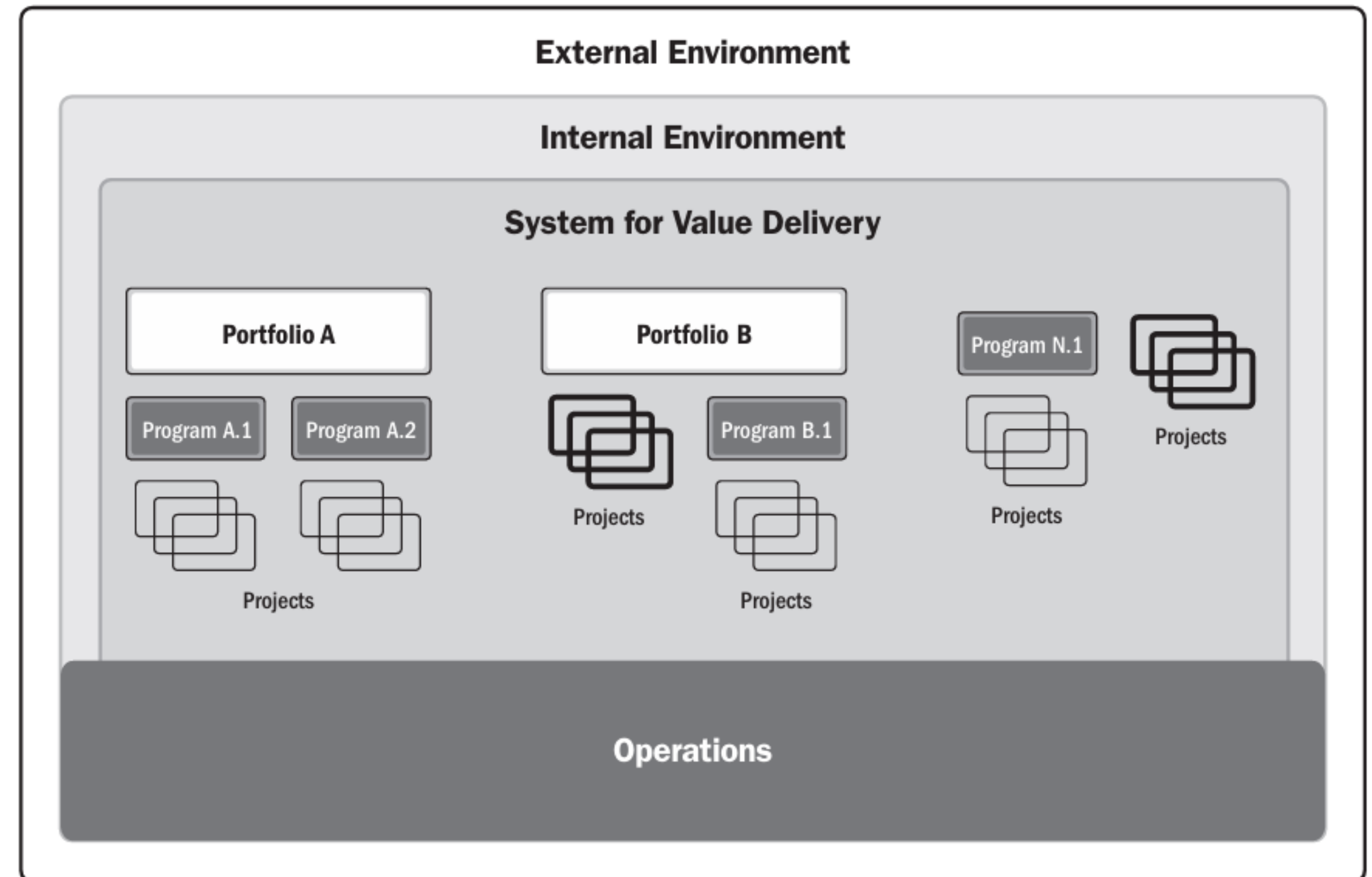


Figure 2-2. Components of a Sample System for Value Delivery

INFORMATION FLOW (PMBOK7)

- From Strategy to Portfolio: Senior leadership sets the strategy, which is translated into priorities and outcomes at the portfolio level.
- From Portfolio to Programs and Projects: Portfolios pass on the desired outcomes, benefits, and value to programs and projects.
- From Programs and Projects to Operations: Programs and projects deliver outputs to operations along with support and maintenance information.
- Reverse Flow: Operations provide feedback for improvements, programs and projects report progress to portfolios, and portfolios share performance evaluations with senior leadership.

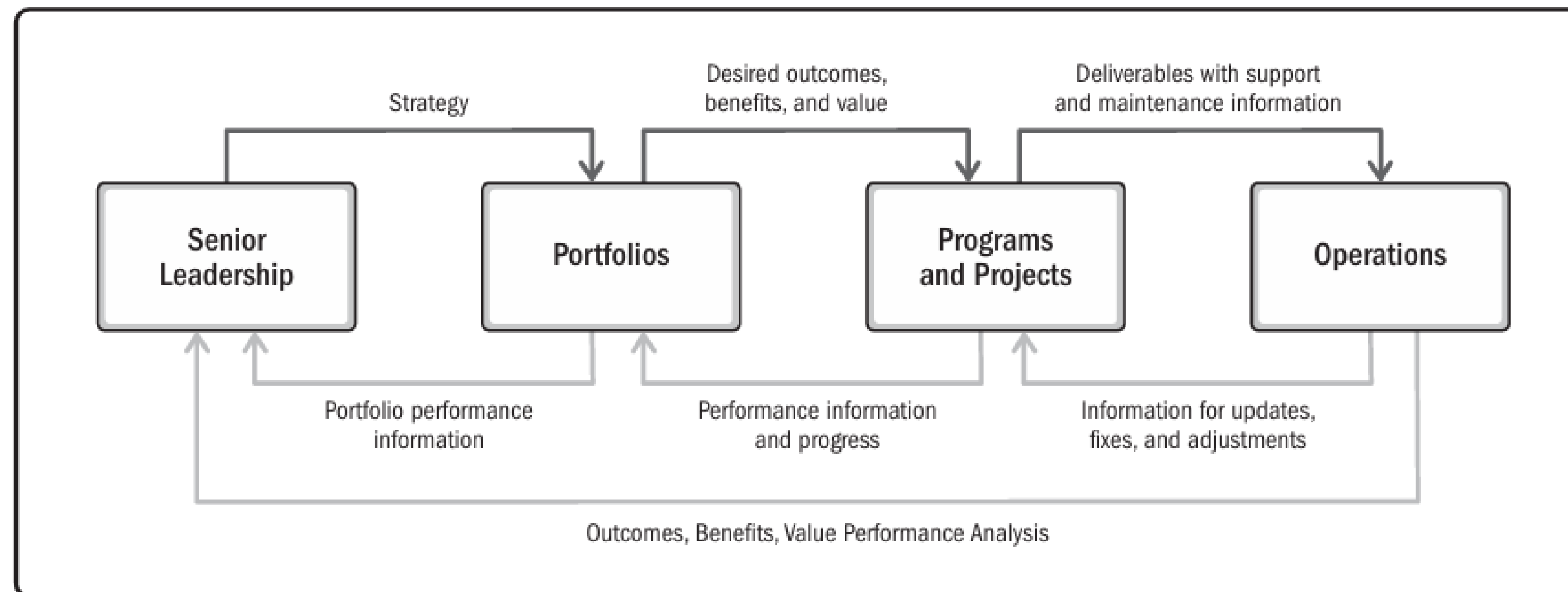


Figure 2-3. Example of Information Flow



The Role of the Project Manager

- Job descriptions vary, but most include responsibilities like **planning, scheduling, coordinating, and working with people** to achieve project goals
- Remember that **97% of successful projects were led by experienced project managers**, who can often help influence success factors





Suggested Skills for Project Managers

- Project managers need a **wide variety of skills**
- They should:
 - Be **comfortable with change**
 - **Understand the organizations** they work in and with
 - Be **able to lead teams** to accomplish project goals





Suggested Skills for Project Managers (cont.)

- They should Understand:
 - The Project Management **Body of Knowledge**
 - Application area **knowledge, standards, and regulations**
 - Project **environment** knowledge
 - **General management** knowledge and skills
 - **Soft skills or human relations** skills



Ten Most Important Skills and Competencies for Project Managers

1. People skills
2. Leadership
3. Listening
4. Integrity, ethical behavior, consistent
5. Strong at building trust
6. Verbal communication
7. Strong at building teams
8. Conflict resolution, conflict management
9. Critical thinking, problem solving
10. Understands, balances priorities



Different Skills Needed in Different Situations

- **Large projects:** leadership, relevant prior experience, planning, people skills, verbal communication, and team-building skills were most important
- **High uncertainty projects:** risk management, expectation management, leadership, people skills, and planning skills were most important
- **Very novel projects:** leadership, people skills, having vision and goals, self confidence, expectations management, and listening skills were most important





Importance of Leadership Skills

- Effective project managers provide leadership by example
- A leader focuses on long-term goals and big-picture objectives while inspiring people to reach those goals
- A manager deals with the day-to-day details of meeting specific goals
- Project managers often take on the role of both leader and manager





Project Management Context





Projects Cannot Be Run in Isolation

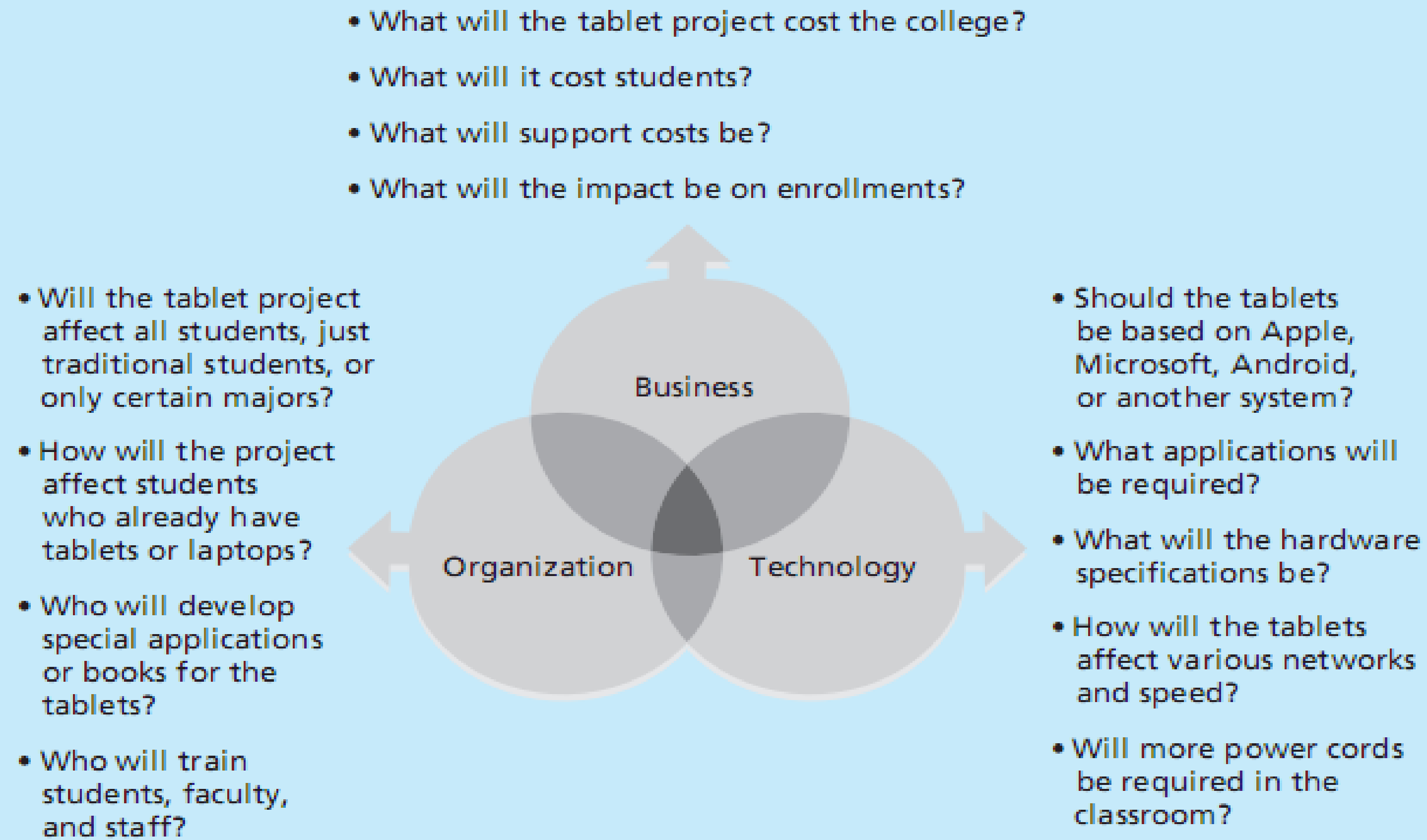
- Projects **must operate in a broad** organizational environment
- Project managers need to use **systems thinking**:
 - Taking a **holistic view** of carrying out projects within the context of the organization
- Senior managers must make sure projects continue to **support current business needs**



A Systems View of Project Management

- A systems approach emerged in the 1950s to describe a more analytical approach to management and problem solving
- Three parts include:
 1. **Systems philosophy**: an overall model for thinking about things as systems
 2. **Systems analysis**: problem-solving approach
 3. **Systems management**: address business, technological, and organizational issues before making changes to systems

Three Sphere Model for Systems Management



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Organizational Structures

Functional: functional managers report to the CEO

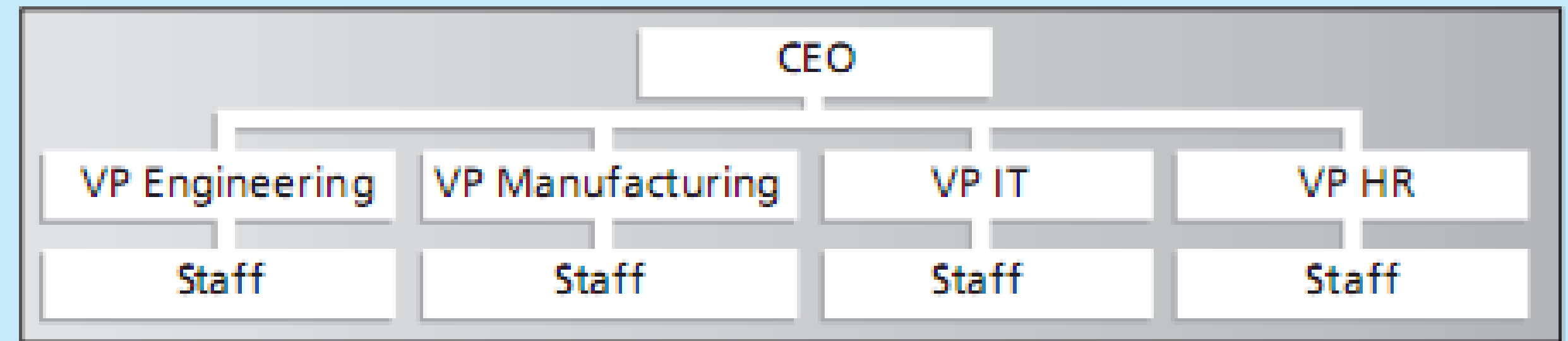
Project: program managers report to the CEO

Matrix: middle ground between functional and project structures; personnel often report to two or more bosses; structure can be weak, balanced, or strong matrix

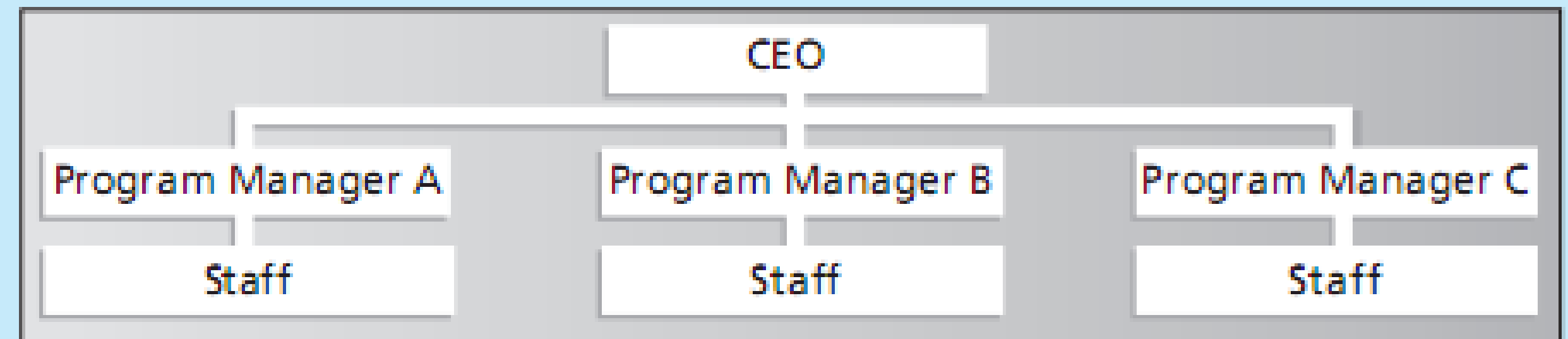


Functional, Project, and Matrix Organizational Structures

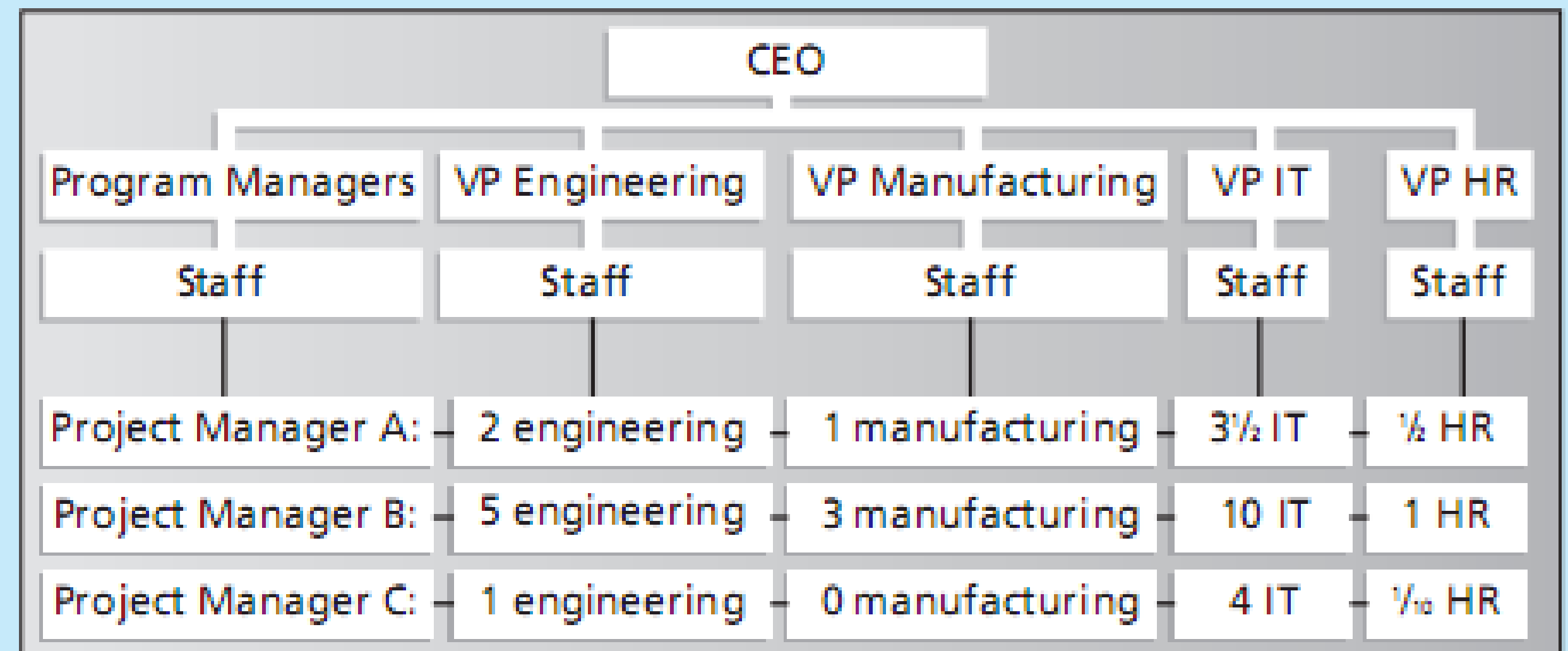
Functional



Project



Matrix



Organizational Structure Influences on Projects

Project Characteristics	Organizational Structure Type				
	Functional	Matrix			Project
		Weak Matrix	Balanced Matrix	Strong Matrix	
Project manager's authority	Little or none	Limited	Low to moderate	Moderate to high	High to almost total
Percent of organization's personnel assigned full-time to project work	Virtually none	0–25%	15–60%	50–95%	85–100%
Who controls the project budget	Functional manager	Functional manager	Mixed	Project manager	Project manager
Project manager's role	Part-time	Part-time	Full-time	Full-time	Full-time
Common title for project manager's role	Project coordinator/ project leader	Project coordinator/ project leader	Project manager/ project officer	Project manager/ program manager	Project manager/ program manager
Project management administrative staff	Part-time	Part-time	Part-time	Full-time	Full-time



Organizational Culture

- Organizational culture is a set of shared **assumptions, values, and behaviors** that characterize the functioning of an organization
- Many experts believe the underlying **causes of many companies' problems** are not the structure or staff, but the **culture**





Ten Characteristics of Organizational Culture

- Member **identity***
- Group emphasis*
- People **focus**
- Unit **integration***
- Control
- **Risk** tolerance*
- **Reward** criteria*
- **Conflict** tolerance*
- Means-ends orientation
- **Open-systems** focus*





Stakeholder Management

- Project managers must take time to identify, understand, and manage **relationships with all project stakeholders**
- Using the **four frames of organizations** can help meet stakeholder needs and expectations
- Senior executives/top management are **very important stakeholders**





The Importance of Top Management Commitment

- People in top management positions are **key stakeholders** in projects
- A very important factor in helping project managers successfully lead projects is the level of **commitment** and **support** they receive from top management
- Without top management commitment, many **projects will fail**
- Some projects have a senior manager called a **champion** who acts as a key proponent for a project





How Top Management Can Help Project Managers

- Providing **adequate resources**
- Approving **unique project** needs in a timely manner
- Getting **cooperation** from other parts of the organization
- **Mentoring** and coaching on leadership issues



Need for Organizational Commitment to Information Technology (IT)

- If the organization has a negative attitude toward IT, it **will be difficult for an IT project to succeed**
- Having a Chief Information Officer (CIO) at a high level in the organization **helps IT projects**
- Assigning non-IT people to IT projects also encourages more commitment



Need for Organizational Standards

- **Standards** and **guidelines** help project managers be more effective
- Senior management can encourage:
 - The **use of standard forms and software** for project management
 - The development and **use of guidelines for writing project plans** or providing status information
 - The creation of a **project management office** or center of excellence



Project Phases and the Project Life Cycle

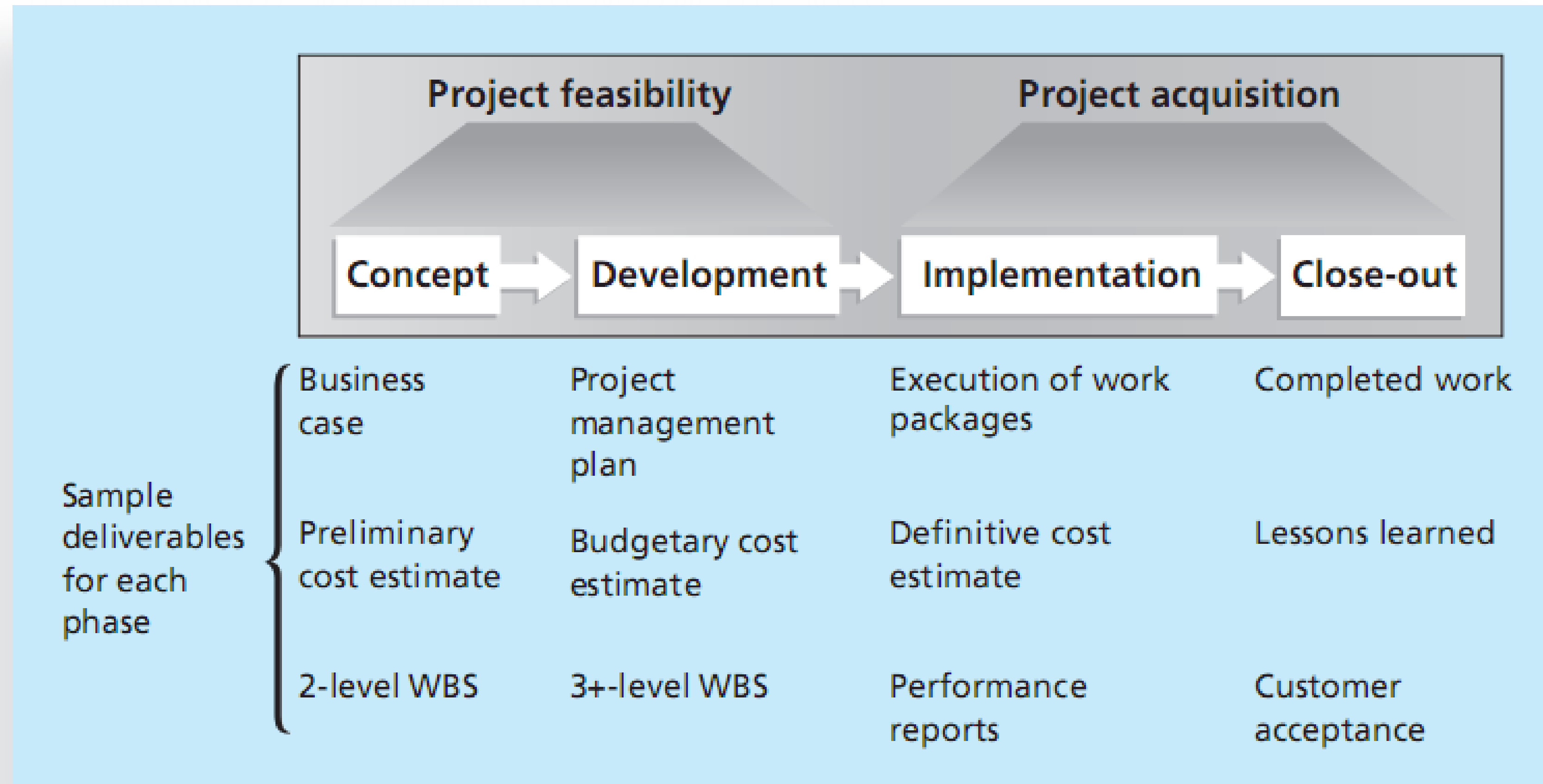
- A **project life cycle** is a collection of project phases that defines:
 - What work will be performed in each phase
 - What deliverables will be produced and when
 - Who is involved in each phase
 - How management will control and approve work produced in each phase
- A **deliverable** is a product or service produced or provided as part of a project



More on Project Phases

- In **early phases** of a project life cycle:
 - Resource needs are usually lowest
 - The level of uncertainty (risk) is highest
 - Project stakeholders have the greatest opportunity to influence the project
- In **middle phases** of a project life cycle:
 - The certainty of completing a project improves
 - More resources are needed
- The **final phase** of a project life cycle focuses on:
 - Ensuring that project requirements were met
 - The sponsor approves completion of the project

Phases of the Traditional Project Life Cycle



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Product Life Cycles

- Products also **have life cycles**
- The Systems Development Life Cycle (SDLC) is a **framework for describing the phases** involved in developing and maintaining information systems
- Systems development projects can follow
 - **Predictive life cycle**: the scope of the project can be clearly articulated and the schedule and cost can be predicted
 - **Adaptive Software Development (ASD)** life cycle: requirements cannot be clearly expressed, projects are mission driven and component based, using time-based cycles to meet target dates





Predictive Life Cycle Models

- **Waterfall** model: has well-defined, linear stages of systems development and support
- **Spiral** model: shows that software is developed using an iterative or spiral approach rather than a linear approach
- **Incremental** build model: provides for progressive development of operational software
- **Prototyping** model: used for developing prototypes to clarify user requirements
- **Rapid Application Development** (RAD) model: used to produce systems quickly without sacrificing quality





Agile Software Development

- Agile software development has become popular to describe new approaches that focus on **close collaboration between programming teams and business experts**





The Importance of Project Phases and Management Reviews

- A project should **successfully pass through each of the project phases** in order to continue on to the next
- Management reviews, also called phase exits or kill points, should occur after each phase to **evaluate the project's progress**, likely success, and continued compatibility with organizational goals



The Context of IT Projects

- IT projects can be very diverse in terms of **size**, **complexity**, **products** produced, **application area**, and **resource** requirements
- IT project team members often **have diverse backgrounds and skill sets**
- IT projects use **diverse technologies** that change rapidly; even within one technology area, people must be highly specialized





Recent Trends Affecting IT Project Management

- **Globalization**: lower trade and political barriers and the digital revolution have made it possible to interact almost instantaneously with billions of other people across the planet
- **Outsourcing**: outsourcing is when an organization acquires goods and/or sources from an outside source; offshoring is sometimes used to describe outsourcing from another country
- **Virtual teams**: a virtual team is a group of individuals who work across time and space using communication technologies
- **Agile** : Method for being able to move quickly and easily



Important Issues and Suggestions Related to Globalization

- **Issues**
 - Communications
 - Trust
 - Common work practices
 - Tools
- **Suggestions**
 - Employ greater project discipline
 - Think global but **act local**
 - Keep **project momentum** going
 - Use **newer tools and technology**



Outsourcing

- Organizations **remain competitive by using outsourcing to their advantage**, such as finding ways to reduce costs
- Their next challenge is to make **strategic IT investments with outsourcing** by improving their enterprise architecture to ensure that IT infrastructure and business processes are integrated and standardized
- Project managers should become more familiar with **negotiating contracts** and other outsourcing issues





Virtual Teams Advantages

- Increasing **competitiveness** and responsiveness by having a team of workers **available 24/7**
- Lowering **costs** because many virtual workers do **not require office space or support** beyond their home offices
- Providing more **expertise and flexibility** by having team **members from across the globe working** any time of day or night
- Increasing the **work/life balance** for team members by **eliminating fixed office hours** and the need to travel to work





Virtual Team Disadvantages

- Isolating team members
- Increasing the potential for communications problems
- Reducing the ability for team members to network and transfer information informally
- Increasing the dependence on technology to accomplish work





Agile Project Management

- Agile means being able to move **quickly** and **easily**, but some people feel that project management, as they have seen it used, **does not allow people to work quickly or easily**.
- Agile today means using a method based on **iterative** and **incremental** development





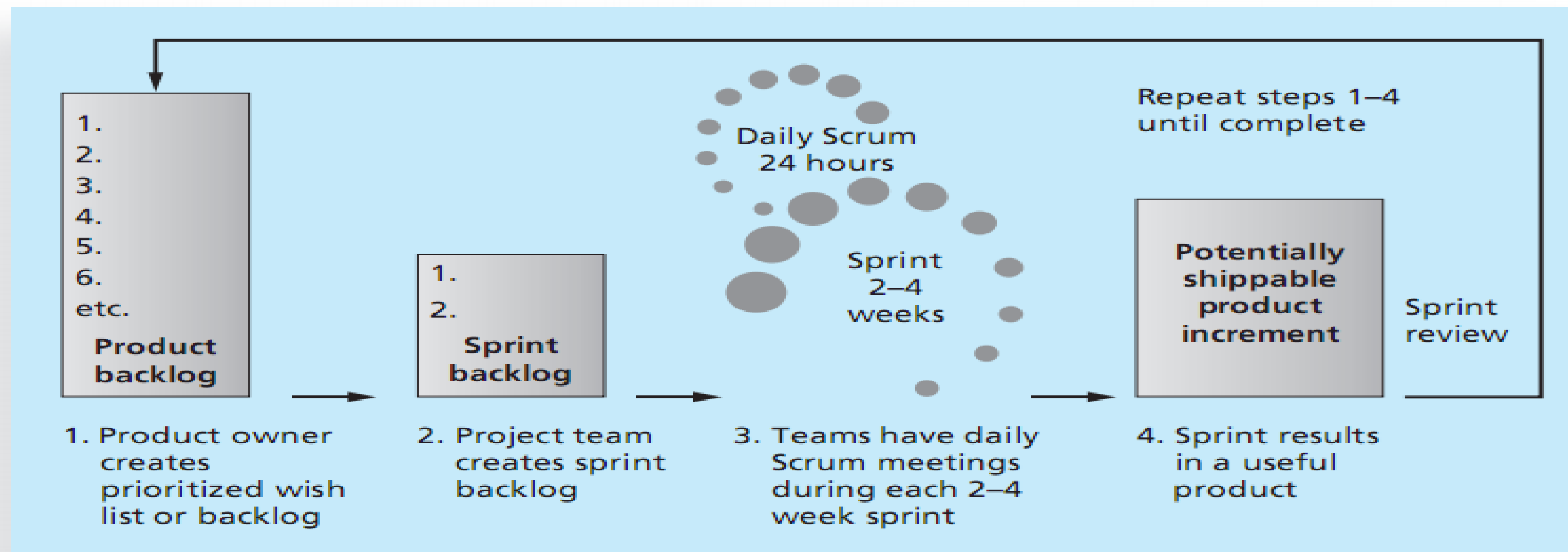
Manifesto for Agile Software Development

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan



SCRUM

- Scrum is the leading agile development method for completing projects with a complex, innovative scope of work



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Agile, the PMBOK® Guide, and a New Certification

- PMI Agile Certified Practitioner (PMI-ACP)
- Recognizes your knowledge of agile principles and your skill with agile techniques
- Organizations that are highly agile and responsive complete more of their projects successfully than their slower-moving counterparts
- Spans many approaches to agile such as Scrum, Kanban, Lean, extreme programming (XP) and test-driven development (TDD.)



References

1. Kathy Schwalbe, *Managing Information Technology Projects 8th Edition*, Course Technology, Cengage Learning, 2018
2. A Guide to the Project Management Body of Knowledge: *PMBOK Guide 6th Edition*, Project Management Institute, 2016